SYSTEM ARCHITECTURE & TECHNICAL IMPLEMENTATION GUIDE

SoleSight Intelligence & RCA Engine

Enterprise Root-Cause Analysis, Causal Decomposition & Decision Intelligence for Retail Footwear Operations

DETERMINISTIC CAUSAL RCA | ZERO LLM HALLUCINATIONS | DUCKDB OLAP ENGINE | FASTAPI + REACT 18 | YAML SEMANTIC CONTRACTS | PYDANTIC FROZEN EVIDENCE | 8-LANGUAGE I18N

DOCUMENTATION ROADMAP & TABLE OF CONTENTS

1. Executive Summary & Solution Vision

SoleSight is an enterprise-grade Root-Cause Analysis (RCA) and Decision Intelligence engine designed for store-based footwear retail chains. Retail leadership is routinely trapped by backward-looking BI dashboards that declare *what* moved (e.g., "Mumbai store conversion plunged -24%") but fail to provide mathematical proof of *why* it moved, whether the diagnosis is statistically trustworthy, and *what exact physical action* should be executed to recover lost revenue.

SoleSight resolves this operational bottleneck by decoupling **statistical computation** from **narrative synthesis**. All causal attribution, signal screening, dose-response regressions, and cross-lagged correlations run strictly deterministically in Python/DuckDB. LLMs are never permitted to guess causes, query arbitrary database tables, or fabricate numbers. Instead, an immutable **Frozen Evidence Object** is passed to persona-tailored synthesis engines that generate actionable, role-authorized recommendations for Store Managers, Regional Directors, CFOs, and Marketing Leads.

NETWORK ANALYZED	RECOVERABLE REVENUE	STATISTICAL CONFIDENCE	REBALANCE ROI
8 Stores / 26 SKUs	**₹54.2 Lakhs**	**$r = 0.783$, $p = 0.02$**	**29.8x**
6-Month Longitudinal Granular History	Network Leakage Attributed & Actionable	Empirical Dose-Response Triangulation	Inter-DC Dispatch Return on Investment

2. The Retail Footwear Problem Space

Footwear retail operations suffer from distinct structural dynamics that generic retail analytics miss:

- **The Size-Curve Stockout Trap:** Footwear demand is non-fungible across sizes. A runner wearing UK 9 cannot purchase UK 6. When core velocity sizes (UK 8 & 9) stock out, store entrance footfall remains high, try-on activity continues, but purchase conversion plummets, creating massive invisible revenue leakage.
- **Compromise Buying & Lagged Returns:** Facing size stockouts, floor associates often nudge shoppers into adjacent sizes (e.g. UK 8.5 or 9.5). This masks conversion loss at the POS register but drives an explosive spike in size-related returns 2 to 5 weeks later.
- **The Multi-Collinearity Blame Game:** When revenue dips, store managers cite staffing shortages, marketing points to adverse weather, and regional planners blame competitor discounts. Traditional analytics cannot isolate the true primary driver.
- **Hallucination Risks in Generic AI:** Standard generative AI tools hallucinate causal links (e.g. claiming "heavy rain reduced store conversion" when rain only impacts entrance traffic). SoleSight eliminates this risk entirely via formal mathematical invariant gates.

3. Solution Approach & Design Principles

Core Architectural Tenets

1. Strict Compute/Narrative Separation: LLMs generate language only. All mathematical operations are 100% deterministic.
2. Honest Uncertainty & Hardcoded Abstention: If candidate drivers lack statistical significance or data is conflicting, the system explicitly abstains from asserting false causality.
3. Semantic Contract Governance: All KPIs are governed by standalone YAML contracts declaring formula, grain, and physical invariants.
4. Feasibility-Checked Actions: Every recommendation is verified against physical supply chain constraints (DC buffer stock, transit lead-times, MOQ).

4. Five-Layer Architectural Blueprint

SoleSight is structured into five isolated, contractually decoupled layers:

```
LAYER 1: DATA INGESTION & TEMPORAL STORAGE (L1)
• In-Memory DuckDB OLAP Engine | Vectorized SQL queries & analytical aggregations
• 12 Relational Tables: fact_pos, fact_inventory, fact_footfall, fact_staff_roster, fact_returns, ext_weather, etc.
• Temporal As-Of alignment, rolling window computation, and zero-copy dataframe exchange

        ▼  DuckDB Analytical SQL & Materialized Metric Frames

LAYER 2: SEMANTIC KPI CONTRACT LAYER (L2)
• Declarative YAML Specifications: conversion_rate, footfall, full_price_sell_through, size_curve_fill_rate, etc.
• Explicit Lineage Trees, Aggregation Rules, and Physical Invariant Boundary Checks [0, 1]

        ▼  Standardized Metric Vectors & Lineage Graphs

LAYER 3: STATISTICAL & CAUSAL RCA ENGINE (L3)
• Stage 1: Rolling 8-Week Z-Score Anomaly Screening (|z| > 2.0σ)
• Stage 2: Dimensional Funnel & Price-Volume-Mix (PVM) Shapley Decomposition
• Stage 3: Causal Triangulation (Dose-Response OLS, Partial Correlation, Lagged Cross-Corr, Weather Residuals)
• Stage 4: Impact Ranking, Feasibility Validation, and P&L Leakage Calculation

        ▼  IMMUTABLE FROZEN EVIDENCE OBJECT (Pydantic Contract JSON)

LAYER 4: AI INSIGHTS & PERSONA SYNTHESIS (L4)
• Zero SQL Execution, Zero Math Computation | Consumes strictly Frozen Evidence JSON
• Persona Engines: Store Ops (floor shifts), Regional (DC transfers), CFO (P&L leakage & ROI), Marketing (campaign ROI)
• Hardcoded Abstention Gate: Blocks AI synthesis if top confidence < 0.50 or drivers conflict

        ▼  REST API Endpoints & Typed JSON Payloads

LAYER 5: MODERN DELIVERY UI & ANALYTICS (L5)
• React 18 + TypeScript + Vite + Tailwind CSS + Recharts
• Dual Experience Modes: SaaS Engineering Dashboard & High-Impact Retail Boardroom
• Interactive AI Copilot, Live Real-Time Telemetry, 8-Language i18n, Offline Resilience
```

5. Semantic KPI Contracts (`backend/contracts/`)

Every operational KPI in SoleSight is defined in a declarative YAML specification to eliminate ambiguity and prevent cross-team metric discrepancy:

Contract File	Cadence	Mathematical Formula	Primary Causal Drivers	Physical Bounds
`conversion_rate.yaml`	Weekly	`POS Transactions / Sensor Footfall`	Size stockouts, Runner assist lag, Fitting queue friction	0.00 ≤ CR ≤ 1.00
`footfall.yaml`	Daily	`Overhead Entrance Sensor Count`	Campaign reach, Weather precipitation, Local events	Count ≥ 0
`full_price_sell_through.yaml`	Monthly	`Units Sold @ ≥ 95% MSRP / Total Inflow`	Markdown depth, Category velocity, Aging inventory	0.00 ≤ ST ≤ 1.00
`size_curve_fill_rate.yaml`	Daily	`Available Core Sizes (UK 6-11) / 6`	DC replenishment cycle, Stock transit lag, Safety buffer	0.00 ≤ FR ≤ 1.00
`size_related_return_rate.yaml`	Monthly	`Size Return Units / POS Units Sold`	Forced compromise sizing, Lagged rack fill-rate deficit	0.00 ≤ RR ≤ 1.00

6. Four-Stage Statistical RCA Pipeline

Stage 1: Signal Detection & Anomaly Screening

To prevent false alarms caused by natural weekly noise, SoleSight computes rolling 8-week trailing statistical baselines per store-SKU:

```
z_score = (Actual_Metric - Rolling_Mean_8w) / Rolling_Std_8w
```

An anomaly is flagged as material (`is_material = True`) if and only if `|z_score| ≥ 2.0σ`. In the hero evaluation window (Week of 2026-06-01), Mumbai Flagship (`STORE-001`) conversion rate dropped from 17.1% to 12.9%, registering an extreme anomaly of **z = -5.44**. Simultaneously, footfall traffic registered **z = -0.25** (within normal baseline), mathematically isolating the failure to the in-store conversion funnel.

Stage 2: Dimensional Funnel & Decomposition

The engine decomposes movements across the organizational tree: Network → Region → Store → Department → Category → SKU → Size. Using Shapley Price-Volume-Mix (PVM) decomposition, it isolates the exact percentage contribution of each dimension. In the hero case, 88.4% of the conversion drop was localized to **Men's Performance Running**, specifically **FW-001 (Marathon Pro)**.

Stage 3: Causal Triangulation Engine

SoleSight rigorously validates causal hypotheses against five empirical statistical tests:

Statistical Test	Hypothesis Evaluated	Empirical Output Across 6-Month Dataset	Assigned Tier
Dose-Response OLS	Stockout Severity vs Conversion Drop	`conv_%_change = -13.94 + 1.329 × units_on_hand` **r = 0.783, p = 0.0215, n = 8 stores**	**HIGH (P=0.92)**
Partial Correlation	Staff Untrained Hours vs Conversion	`r = -0.107, p = 0.596` (controlling for stock) Identifies grain mismatch (monthly roster vs weekly POS).	**LOW (P=0.35)**
Lagged Cross-Correlation	Stock Deficit Predicting Returns	Correlation flips from positive at Lag 0 to negative at **Lag 2 (r = -0.379)** and **Lag 5 (r = -0.447)**.	**MEDIUM (P=0.68)**
Weather Residual OLS	Monsoon Rain Impact on Footfall	`Footfall = 352.4 - 1.21×Precip + 117.18×Weekend` **R^2 = 0.314, p < 0.001**. Explains late June traffic dip.	**HIGH (P=0.88)**
Event Lift Analysis	Marathon / City Festival Lift	**+58.1% Footfall lift** during marathon event window. Correctly isolates temporary demand surge.	**HIGH (P=0.85)**

Stage 4: Impact Ranking, Feasibility & Action Synthesis

The engine computes net recoverable revenue leakage using the counterfactual lost margin:

```
Recoverable_Leakage = Footfall × (Target_CR - Actual_CR) × Average_Order_Value × Gross_Margin%
```

For STORE-001, this yields **₹13.4 Lakhs**. The system verifies DC inventory levels and dispatch truck schedules, synthesizing a concrete recommendation: "*Emergency dispatch 48 units of Marathon Pro (Sizes UK 8 & 9) from Bhiwandi Central DC (ETA 18 hrs, cost ₹45,000, ROI 29.8x)*".

7. Hardcoded Abstention Gate & Confidence Calibration

SoleSight guarantees absolute truthfulness through two built-in statistical safety mechanisms:

The Abstention Gate (PRD §5)

When all candidate drivers score in the `LOW` confidence tier (e.g. Conflicting signals between weather, competitor discounts, and fitting queue friction where no driver exceeds P ≥ 0.50), the system **hard-abstains** from asserting a false cause. It outputs an honest disclaimer and recommends physical floor audit verification.

Sparse-History Calibration (PRD §6)

For newly launched SKUs (e.g. `SKU-9901 Trailblazer`) with fewer than 3 weekly observations, historical variance cannot be established. The engine sets `is_sparse_history: true` and widens tolerance bands to ±4.5σ, suppressing false-positive panic alarms while baseline forms.

8. Persona-Tailored Decision Intelligence & RBAC

Different retail leaders operate under different decision rights. SoleSight generates four tailored operational lenses from the same underlying Frozen Evidence Object:

Role & Persona	Assigned Scope	Decision Rights	Tailored Narrative & Action Focus
Store Operations Manager `Rahul Sharma`	`STORE-001` (Mumbai Flagship)	Floor shifts, stockroom runner allocation, floor re-merchandising.	Floor try-on drop-offs, core size stockout (UK 8/9), runner assist lag, request DC stock priority.
Regional Operations Director `Vikram Malhotra`	`West Region` (Mumbai, Pune, Ahmedabad)	Inter-store transfers, DC logistics, regional safety buffer.	Inter-branch inventory reallocations from Pune (overstocked) to Mumbai, regional benchmark comparison.
CFO & Finance Director `Priya Mehta`	`Enterprise Network`	Working capital, emergency logistics budget, pricing policy.	Recoverable revenue leakage (₹54.2L network / ₹13.4L store), margin erosion, dispatch ROI (29.8x).
Marketing & Growth Lead `Ananya Sen`	`Acquisition & Campaigns`	Campaign budget, channel allocation, promo timing.	Ad spend attribution, Nitro campaign footfall lift (+3.9%) vs store-level stockout conversion bottleneck.

> **Server-Side Security Enforcement (PRD §7)**
> Role boundaries are enforced on the FastAPI backend (`backend/app/core/security.py`). If a Store Manager attempts to query foreign store data, the server returns an immediate `HTTP 403 Forbidden`, preventing cross-tenant data leakage.

9. Key Evaluator Demo Scenarios (1-Click Triggers)

The application includes 1-click scenario triggers in the top navigation bar:

- **Scenario A: Multi-Factor Stockout Hero (STORE-001):** Footfall +3.9% from campaign, conversion drops -24.0% due to UK 8 & 9 stockout on Marathon Pro ($r=0.783$, $p=0.02$). Quantifies ₹13.4 Lakhs leakage with 29.8x ROI action.
- **Scenario B: Contradictory Low-Confidence Abstention:** Low confidence across all drivers ($P=0.40$). Triggers hardcoded abstention gate, honestly refusing to guess and advising physical floor audit.
- **Scenario C: Sparse-History SKU Calibration (SKU-9901):** New SKU with only 2 weeks of history. System expands tolerance bands to $\pm 4.5\sigma$ and marks `is_sparse_history: true` to avoid false alarms.
- **Scenario D: Monsoon Weather Traffic Anomaly:** Week of June 29 footfall drops to $z = -2.45$. Engine attributes dip to heavy rainfall ($p < 0.001$), correctly separating weather traffic loss from conversion-side stockouts.

10. Modern Delivery UI & Interactive UX

The frontend is built with React 18, TypeScript, Vite, Tailwind CSS, and Recharts:

- **Dual Experience Switcher:** Instantly toggle between **SaaS Engineering Dashboard** (deep statistical charts, correlation scatterplots, dose-response curves) and **Retail Boardroom Mode** (executive presentation with audio narration, KPI scorecard, and high-level decisions).
- **Interactive AI Copilot Drawer:** Domain-aware assistant allowing operators to ask ad-hoc questions (e.g. "*Why did conversion drop in Pune?*", "*Simulate dispatching 50 units*") with real-time email dispatch modal integration.
- **Real-Time Telemetry & Model Calibration:** Inspect pipeline latency, token usage, Pydantic validation status, and adjust causal significance thresholds live.
- **8-Language Internationalization (i18n):** Full localized support for English, Hindi (हिंदी), Marathi (मराठी), Gujarati (ગુજરાતી), Tamil (தமிழ்), Telugu (తెలుగు), Kannada (ಕನ್ನಡ), and Bengali (বাংলা).
- **Zero-Friction Offline Demo Mode:** Automatically activates pre-cached dataset if backend is unreachable, guaranteeing 100% interactive presentation reliability.

11. Backend API Reference & Pydantic Contracts

Endpoint	Method	Parameters / Request Body	Output Description
`/api/v1/kpi/evaluate`	`GET`	`kpi_id, store_id, period, region`	Evaluates KPI values, rolling baselines, and z-score anomaly status.
`/api/v1/evidence/explore`	`GET`	`kpi_id, store_id, period`	Returns complete Frozen Evidence Object with all hypotheses, tiers & regression stats.
`/api/v1/narrative/synthesize`	`POST`	`FrozenEvidencePayload, persona`	Synthesizes persona-tailored narrative and feasibility-checked action plan.
`/api/v1/copilot/chat`	`POST`	`message, persona, store_id, context`	Interactive Copilot answering operational queries and generating dispatch drafts.
`/api/v1/constraints/validate`	`POST`	`action_id, store_id, units`	Validates DC inventory buffers, truck transit times, and minimum order quantities.
`/api/v1/feedback/submit`	`POST`	`evidence_id, rating, comments`	Logs operator feedback for continuous RCA model calibration.

The Frozen Evidence Object Contract (Pydantic Schema)

```
class FrozenEvidenceObject(BaseModel):
    evidence_id: str
    kpi_id: str
    store_id: str
    period: str
    actual_value: float
    baseline_value: float
    z_score: float
    is_material: bool
    is_sparse_history: bool
    abstain: bool
    abstain_reason: Optional[str] = None
    hypotheses: List[CausalHypothesis]  # driver, score, tier, regression_stats
    counterfactual_leakage_inr: float
    top_ranked_action: ActionRecommendation  # action, cost, roi, feasibility_checked
```

12. Automated Verification & Quick Start Guide

Unified 1-Command Launcher

```
python run.py
```

Boots FastAPI on port 8000, Vite on port 5173, checks server readiness, and opens the browser dashboard automatically.

Automated Verification Test Suite

```
python backend/verify_all.py
```

Executes 5 automated validation tests covering all 5 YAML contracts, DuckDB in-memory ingestion, the Hero Multi-Factor stockout scenario, the Low-Confidence Abstention gate, Sparse-History calibration, and server-side RBAC boundary security.

> **System Verification Result**
> [PASS] 5/5 YAML Semantic Contracts Validated
> [PASS] In-Memory DuckDB Engine Initialized across 12 Relational Tables
> [PASS] Hero Multi-Factor Stockout Attribution Verified (Signal: z = -5.44, Top Driver: Size Stockout, Conf = 0.92)
> [PASS] Low-Confidence Abstention Gate & Sparse-History Filters Verified
> [PASS] Role-Based Server-Side Access Control Verified (HTTP 403 Cross-Store Protection Active)